



The (dis)connection between R&D and productivity in China: Policy implications of R&D tax credits

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ABSTRACT

We use Chinese firm-level data from 2001 to 2007 to estimate the (dis)connection between firms' R&D and productivity and find that the productivity effect of R&D investment is very low. We conjecture that firms conduct/report unproductive R&D in order to obtain policy benefits. To explore this plausible misconduct, we investigate the effects of China's 2003 R&D tax reform on firms' R&D investment. The reform generates exogenous treatment variations across firm ownerships for causal identification. We find that the reform has statistically significant and positive effects on firms' R&D investments. Quantitatively, the reform raises firms' R&D investments by 6.68 % and the estimated R&D elasticity of tax deduction for private firms is 0.9147, which is comparable to other countries. However, our further empirical results indicate that the policy-induced R&D is less efficient in promoting firms' productivity than the spontaneous R&D. We provide evidence of firms' relabeling non-R&D expenses to R&D expenses, which partly explains the inefficiency of R&D investment.

1. Introduction

The rapid and sustained economic growth of China since 1978 has transformed one of the poorest countries to an upper-middle-income and the second largest economy in the world. Productivity has been found to be one of the most important factors contributing to this “miracle”. Zhu (2012) estimates that the total factor productivity (TFP) grew at 3.16 % during 1978–2007, accounting for 78 % of the growth in GDP per capita.² The productivity growth was more pronounced during 1998–2007, at 4.68 % (Zhu, 2012), and higher in the manufacturing sector at 13.4 % (Brandt et al., 2012). Around the same time, R&D investments also surged in China. According to the World Bank data, China's R&D to GDP ratio jumped from 0.56 % in 2001 to 2.13 % in 2017 at a level comparable to many developed countries.³ As for the level of R&D investment, Fig. 1 shows that China increased R&D investments rapidly after 2000, and reached a level close to that of the United States in 2016, much larger than many developed countries/regions. A natural question emerges: do these R&D investments contribute to the productivity growth in China?

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² Bosworth and Collins (2008), Perkins and Rawski (2008), and Brandt and Zhu (2010) report similar results.

³ See <https://data.worldbank.org/indicator/GB.XPD.RSDV.GD.ZS?locations=CN>.

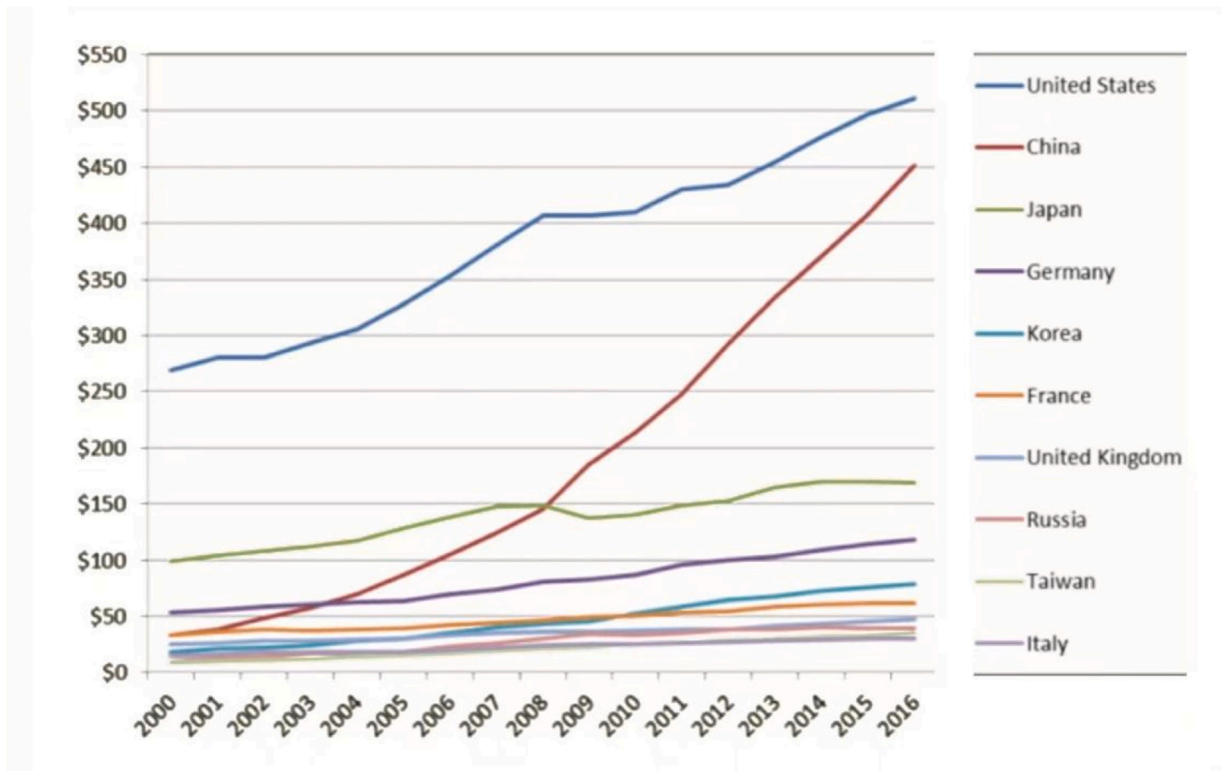


Fig. 1. R&D expenditures of selected countries/regions (2000–2016).

(in billions of current PPP dollars)

Source: CRS analysis of organization for economic cooperation and development, OECD stats database, https://stats.oecd.org/Index.aspx?DataSetCode=MSTI_PUB.

In this paper, we first apply the estimation method of Aw et al. (2011) to Chinese manufacturing firms in 2001–2007 and find that the R&D investment is *not* effective in improving firm-level productivity.⁴ A 1 % increase in R&D investments on average leads to a 0.00000536 % increase in firm-level productivity. For comparison, taking the electronic industry as an example, the productivity elasticity of R&D in mainland China is 0.0001373 whereas that in Taiwan is 0.0067 (column 2 in Table 2, Aw et al., 2011), which is around 50 times larger than the former. These results indicate the inefficiency of R&D on productivity in mainland China. This disconnection between firm-level R&D investment and productivity is puzzling, especially when we compare it with the aforementioned co-movement between R&D and productivity growth at the macro level. The ensuing question is what drives this disconnection.

A few explanations have been provided or mentioned by previous research. One explanation is related to the distance to the technology frontier. König et al. (2022) find that R&D investment in China played a less important role as a driver of productivity growth in the 2001–2007 period than in the 2007–2012 period. As Chinese firms were further away from the technology frontier in the earlier period than in the later period, R&D investment was less efficient in the earlier period than in the later one. This argument is consistent with Acemoglu et al. (2006) prediction that innovation is more important for countries that are closer to the technology frontier. Another explanation is related to resource misallocation. König et al. (2022) find that resource misallocation reduced R&D efficiency even in the 2007–2012 period because it encouraged the wrong type of firms (the less efficient ones) to conduct R&D investment. The third candidate is firms' mis-reporting of R&D expenditure. Chen et al. (2021) find evidence of mis-reporting by certain Chinese firms, from 2008 to 2011, who manage to be qualified for the InnoCom program to receive a preferential tax rate. They suggest that the mis-reported R&D is unproductive. Our paper explicitly addresses this last factor by examining the R&D-inducing effects of the most comprehensive R&D tax credit policy in China.

It is well known that the Chinese government plays an active role in promoting innovation (Ding and Li, 2015; Zilibotti, 2017; Chen et al., 2021; König et al., 2022). The government introduces a series of innovation incentive policies in different periods and sectors. The most comprehensive one is the corporate income tax (CIT) incentive program. The program was introduced in 1993 and has been

⁴ Specifically, we first structurally estimate the productivity taking into account of the endogenous R&D and exporting decision according to the model in Aw et al. (2011). Then we apply the data to Equation (8) in Aw et al. (2011) to estimate the effect of Chinese firm-level R&D investments on productivity between 2001 and 2007. In this equation, the coefficient α_4 is our focus of interest because it captures the direct effect of R&D investments on firm-level productivity.

in existence since then. According to the program, 100 % of a firm's R&D expenses can be deducted from the firm's CIT payable. From 1996, an additional 50 % of the R&D expenses can be deducted from a firm's CIT payable if the firm is a state-owned enterprise (SOE) or collectively owned enterprise (COE). Since 2003, the government has extended the policy coverage to all firms, i.e., including private enterprises. We must address two issues in order to see how this CIT incentive program can help explain the disconnection between R&D investment and productivity. First, is the program effective in promoting firms' R&D investment? Second, are all R&D investments real, i.e., are they all true investments taken by the firms to raise productivity?

The distinct features of China's CIT incentive program together with the firm-level R&D data from the Above-Scale Industrial Firms Panel (ASIFP) dataset allow us to answer the above two questions. On the one hand, the policy revision in 2003, which we refer to as the *2003 R&D reform*, eliminates the program's discriminatory nature, which is based on pre-determined registered type, i.e., covering only SOEs and COEs before 2003. This policy change enables us to circumvent the most common and difficult obstacles in estimating the effects of tax policies, which are usually applied to all firms uniformly or based on endogenous historical performance related to the firms' R&D.⁵

Specifically, we conduct our analysis based on Chinese manufacturing firms from 2001 to 2007. We focus on the 2003 R&D reform and conduct a solid differences-in-differences (DID) analysis of its effect on firms' R&D investments. We take SOEs and COEs, who have been covered by the policy since 1996, as the control group, and the other firms (mainly domestic private firms), who have been covered by the policy since 2003, as the treatment group.⁶ Our results show that the 2003 R&D reform has statistically significant and positive effects on the treated firms' R&D investment. Quantitatively, the estimated R&D elasticity of CIT deduction for the treated firms is 0.9147, which is comparable to estimates in the literature.⁷

A critical identification assumption is that conditioning on all the control variables, the R&D behavior shall be comparable across the treatment and control groups before the reform. However, since the control group mainly consists of SOEs and COEs, while the treatment group consists of private firms, they may have different trends before the reform, which makes them incomparable. To alleviate this concern, we conduct a series of tests to confirm the comparability. First, we run a flexible estimation and show that the treatment and control groups do have comparable pre-trends. Second, to tease out potential effects from SOEs, which have different incentive schemes and therefore may have different pre-trends with private firms, we further include the SOEs' specific trends in our regression. The results remain consistent. Third, COEs are regarded similar to private firms in management and operation (Zhou et al., 2017; Du and Girma, 2009). We keep only COEs as our control group to investigate the policy's effect. We find that COEs do also have parallel pre-trends with the treated firms and the estimated average treatment effect remains consistent. Fourth, we use Propensity Score Matching (PSM) method to construct an alternative control group to re-estimate the effect. The two groups present parallel pre-trends and the baseline finding is robust as well.

Our finding is also robust to various other concerns such as the WTO entry, a bunch of contemporaneous policy changes, and alternative samples. Moreover, inspired by a series of studies that emphasize the importance of firm characteristics, such as ownership, profit, and productivity, in affecting firms' innovation behaviors (e.g. Hall and Van Reenen, 2000; Choi et al., 2011; Howell, 2016; Zwick and Mahon, 2017; König et al., 2022; Cheng et al., 2019), our study explores a set of heterogeneous effects of the 2003 R&D reform. For example, we find that the effects are more pronounced for firms with higher profits and higher productivity.

Although the effect of the 2003 R&D reform on R&D investments is comparable to estimates in other countries, the policy's real effect, i.e., effect on firms' productivity, is unknown. We estimate the elasticity of R&D on firm productivity following Aw et al. (2011) and find that the elasticity of R&D on productivity has decreased from the pre-reform to post-reform periods for the treated firms, but not for firms in the control group. The result suggests that the policy-induced R&D is not as effective as the spontaneous R&D in promoting productivity. This result cannot be explained by the distance to the technology frontier or the misallocation argument.⁸ Instead, we provide an alternative explanation: fake R&D investment. The 2003 R&D reform specifies conditions for receiving tax credits. Firms may manipulate their R&D expenses, for example, relabeling some non-R&D to R&D expenses, in order to meet the conditions. Relying on various features of the reform, we find convincing evidence of firms' mis-reporting and relabeling of their R&D expenditure. For example, the policy was announced on November 27 in 2003, but the qualified firms can apply for a tax deduction on all R&D expenses of the whole year. We do not expect the firms' real R&D investments to adjust to the new policy in a short period (around one month). However, we find that compared to 2002, firms that need to manage their R&D investments to meet the conditions have (1) a sudden increase in R&D expenses, (2) a significant drop in non-R&D administrative expenses,⁹ but (3) no change in total sales. We also find that the productivity effect of R&D is insignificant for firms that have stronger incentives of relabeling.

In sum, we show that although the 2003 R&D policy is effective in stimulating R&D expenses in China, the increased R&D is not very effective in promoting firm productivity. One reason is that firms report fake R&D investments to obtain tax credits. Although our findings are based on the treated firms, i.e., the domestic private firms, after 2003, it is reasonable to expect a similar conclusion can

⁵ As explained by Kasahara et al. (2014), "Estimating the effect of R&D tax credit policy is often difficult because, typically, the same R&D tax credit rate uniformly applies to all firms, and hence, there is no variation across firms to identify the effect of R&D tax credit policy on R&D expenditure," and by Thomson (2017), "Since tax policy treats all equivalent firms the same way, variation in firms' effective benefits from tax credits arises entirely from differences between firms—typically differences in profit status and historic R&D expenditure. Unfortunately, from a statistical perspective, these differences are endogenous to the current level of R&D investment."

⁶ We do not include foreign-invested firms in our sample as they might have very different R&D incentives.

⁷ See reviews by Hall and Van Reenen (2000) and Kasahara et al. (2014) for elasticities of R&D policies in the literature.

⁸ The lower efficacy could also be due to decreasing marginal return of R&D investment, which is hard to test though.

⁹ In China, R&D expenditure is part of the total administrative expenses.

also be made for SOEs and COEs even before 2003 as the same incentives apply to them since 1996. Thus, the aforementioned disconnection between R&D investments and productivity is a result from all firms.

Our paper contributes to the empirical literature of fiscal incentives for innovation and their effects on firm performance (Klette et al., 2000; Czarnitzki et al., 2011; Cappelen et al., 2012; Guo et al., 2016). The first strand of the literature is about the effects of incentive policies on raising R&D investments (González and Pazó, 2008; Hud and Hussinger, 2015). Hall and Van Reenen (2000) provide a review of earlier studies. Despite the unsatisfactory methodologies used in some of those studies, they conclude that a dollar in tax credit for R&D stimulates a dollar of additional R&D, which is comparable to our estimate. Bloom et al. (2002) conduct a comprehensive study on various tax incentives in nine OECD countries over 19 years (1979–1997). By using cross-country panel data, they find positive effects of fiscal incentives on R&D investment. Kasahara et al. (2014) provide a brief update of the literature after Hall and Van Reenen (2000) and conclude that the recent findings support the conclusion by Hall and Van Reenen (2000). In addition, Kasahara et al. (2014) examine the Japanese R&D tax credit reform in 2003 and find that the reform has a positive and significant influence on firms' R&D expenditure with a relatively larger influence on firms with higher outstanding debts. Bøler et al. (2015) explore the effect of the tax credit policy reform of R&D expenses in Norway in 2002 on firms' R&D and international sourcing behaviors. The policy allows firms to deduct 20 % of their R&D expenses from their tax payables. Dechezleprêtre et al. (2016) analyze a policy reform of the UK in 2008 that raises the size threshold (defined by employment and sales) under which small- and medium-sized enterprises (SMEs) firms can access a more generous tax regime, namely, the additional enhancement rate of 50 % of qualifying R&D expenditure from taxable profits on top of the 100 % deduction that applies to any form of current expenditure that is available to all firms.¹⁰

Although we can view our study simply as one of the many in this strand of literature and our study finds similar positive effects of R&D credit on firms' R&D investments (Stoneman, 1991; Yang et al., 2012), China's 2003 R&D reform provides some advantages over those mentioned above in conducting a DID analysis of the policies. For example, in Kasahara et al. (2014), all firms face the same policy, but the researchers are still able to use a DID analysis because firms need to satisfy certain conditions to be qualified for additional credit, which enables the researchers to build their treatment and control groups. The same technique is adopted by Bøler et al. (2015) and others. However, this division of treatment and control groups is somewhat endogenous because it is subject to possible manipulation by firms around the threshold of the division. In the Japanese 2003 tax credit reform, firms can apply for the tax credit only if R&D expenditure in the current fiscal year is greater than the average of the three largest yearly R&D expenditures from the previous five years.¹¹ Therefore, a firm has room to manipulate its R&D expenditures over time to become qualified for the policy. In contrast, in China's 2003 R&D reform, the division of the treatment and control groups is strictly exogenous based on the pre-registered type and not on previous firm performances.

Bloom et al. (2002, p.3) point out that "Generalizing results obtained from using the data on one country to other countries can be misleading." This warning is especially important when we want to draw lessons from developed countries' experiences for developing countries. However, studies based on data from developing countries are extremely scant. Recently, several scholars have started to investigate the case of China.¹² China has introduced several tax credit policies to stimulate innovation in the past three decades. Chen et al. (2021) consider the policy that grants a preferential tax rate (15 % as opposed to the standard 25 %) for China's InnoCom program on qualified companies in China from 2008 to 2011. They focus on the local notching effects induced by the program and find significant increases in those companies' reported R&D investments. Jia and Ma (2017) explore the same policy as Chen et al. (2021) but focus on the listed companies in China from 2007 to 2013. They also find significant increases in those companies' R&D expenditures.¹³ The InnoCom program analyzed by these authors is a special innovation policy because it is available only for new and high-tech enterprises (NHTEs), which are selective and on average very productive. A key qualification requirement of the InnoCom program is R&D intensity. The tax deduction also does not depend on the amount of R&D expenditure. By contrast, the 2003 R&D reform that we study in this paper is a more general R&D policy and the effective tax rate depends completely on a firm's R&D expenditure.

The second strand of the literature is about the effects of R&D investments on productivity. In this regard, our work is complementary to König et al. (2022), which also investigates the effect of R&D on productivity in China. While they emphasize the

¹⁰ Rao (2016) finds a positive impact of an R&D tax credit between 1981–1991 in the US. Unlike most other countries, the US policy, which was introduced in 1981, is incremental in nature and aims to reward R&D expenses in excess of what the firm would have spent in the absence of the credit. Mukherjee et al. (2017) also study the effects of the US R&D tax credit.

¹¹ According to Bøler et al. (2015), the key qualification criterion in Norway's R&D policy reform in 2002 is that the firm's project is less than NOK 4 million R&D expenditure (US\$ 0.5 million). According to Dechezleprêtre et al. (2016), the UK policy gives additional credits to SMEs and there is a threshold to define SMEs.

¹² Ivus et al. (2021) study the effects of an R&D tax credit on innovation in India and find strong positive effects on firm R&D expenditures, R&D intensity, and the number of patent applications.

¹³ Cai et al. (2018) examine the impacts of a tax reform implemented in 2002 that switches the CIT collector, from the local tax bureau to the state tax bureau, of all manufacturing firms established in or after January 2002. The reform of the collection method results in a 10% reduction in the effective tax rate for firms and they find that the reform improves firm innovation in both quantity and quality. Shao and Xiao (2019) focus on China's 2006 tax base reform, which doubles the wage deduction for the domestic firms in their CIT calculation from 800RMB per worker per month to 1600RMB per worker per month. They argue that the reform leads to a significant increase in firms' patent applications. However, there are key differences between the type of policy we study and those examined by these two papers. The 2002 reform analyzed by Cai et al. (2018) and the 2006 reform by Shao and Xiao (2019) are general CIT policies, that is, not directly designed to encourage firms' innovation, such as R&D investments. By contrast, the 2003-R&D reform is a pure R&D policy.

misallocation of R&D investments across firms in explaining the aggregate productivity, our paper focuses on the within-firm productivity effect of the firm's R&D, identified by a quasi-exogenous policy shock. In addition, their paper takes the misallocation as given, and our paper identifies an important source of misallocation of R&D: the policy-induced fake R&D investments. Like us, [Chen et al. \(2021\)](#) also find evidence of relabeling. However, the two papers have significant differences. First, [Chen et al. \(2021\)](#) focus on the InnoCom program in China from 2008 to 2011 to show that firms have incentives to relabel some non-R&D expenditures as R&D expenditures to qualify for the program and obtain tax credits. We provide evidence of relabeling under the more comprehensive program, i.e., the 2003 R&D reform, during a different period, 2001–2007. Second, considering the relabeling behavior, [Chen et al. \(2021\)](#) build a structural model to estimate the effect of real R&D on productivity growth and draw policy and welfare implications. Our study focuses on identifying the R&D promoting effect of the 2003 R&D reform. As a result, our logic is the following: large-scale R&D tax credit policies effectively stimulate reported R&D, there is clear evidence on relabeling of R&D expenses in response to policies, and therefore the overall R&D efficiency is low in promoting productivity.

The rest of the paper is organized as follows. [Section 2](#) describes institutional background and data. [Section 3](#) establishes the fact that the effect of R&D on productivity in China during 2001–2007 is quantitatively small, and we propose an explanation for this low effect, that is, the R&D may be distorted by government's R&D promotion policies. To check this possibility, in [Sections 4 to 6](#) we explore the role of the 2003 R&D reform in explaining this disconnection between R&D and productivity. Specifically, [Section 4](#) investigates the effects of the 2003 R&D reform on firms' R&D, to show that the reform is effective in inducing firms' R&D. [Section 5](#) analyzes the real effects of the 2003 R&D reform on firm-level productivity to show that the policy-induced R&D is less effective than spontaneous R&D in promoting productivity. [Section 6](#) demonstrates evidence of policy-induced relabeling of R&D, which provides an explanation to the inefficiency of R&D. [Section 7](#) concludes the paper.

2. Background and data

2.1. CIT incentive program

The CIT system of China was stable during 2000–2007. The CIT was on a dual-track scheme with a base tax rate of 33 % for all domestic-owned enterprises (DOE) and a preferential rate ranging from 15 % to 24 % for FIEs. In 2008, the Chinese government implemented a major corporate tax reform that eliminated the dual-track system and established a common tax rate of 25 % for all enterprises ([Chen et al., 2021](#)).

To incentivize R&D investment, China has implemented a large-scale deduction tax policy of enterprises' R&D expenses since 1993. The policy regulates that 100 % of firms' R&D expenses can be deducted from CIT payables for all firms.¹⁴ Firms are required to submit their R&D project plans *ex-ante* to the local Science and Technology Bureau and the local tax bureau to obtain the deduction approval. Firms also need to report their actual R&D expenses at the end of each accounting year. In 1996, the Chinese government introduced an additional 50 % deduction of R&D expenses from CIT payables to a favored group of enterprises.¹⁵ If firms in this group maintained their R&D growth rates no less than 10 % in two consecutive years, they are eligible for the application of additional deduction of R&D expenses. This favored group consists of SOEs (registration type 110), COEs (registration type 120), joint-stock enterprises (registration type 130) controlled by state-owned or collective capital, state-owned joint-venture enterprises (registration type 141), collective joint-venture enterprises (registration type 142), state-owned and collective joint-venture enterprises (registration type 143), and other joint-venture enterprises (registration type 149) controlled by state-owned or collective capital. The group, which is composed mainly by SOEs and COEs, has benefited from this policy since 1996.

To promote fair competition and improve economic efficiency among all enterprises, the Ministry of Finance and the National Tax Bureau of China implemented a major revision to the 1996 CIT additional deduction policy on November 27, 2003. This revision applied the additional 50 % deduction to all manufacturing enterprises, including domestic enterprises and FIEs.¹⁶ Therefore, while SOEs and COEs continued to enjoy this policy, domestic private firms and FIEs began to enjoy this additional deduction of 50 % of R&D expenses from CIT payables in 2003.

The preferential tax arrangement provides economically nontrivial incentives to eligible firms. The Chinese CIT rate is 33 % for domestic firms, and thus, the additional 50 % deduction of R&D expenses from tax payable means a 16.5 % of R&D expenses reduction in cost. In addition to the original 100 % deduction, the total deduction is 150 %, which means a total of 49.5 % of R&D expenses reduction in cost.¹⁷

During the same period, China also implemented many other CIT-related policies and reforms, such as preferential CIT policy designed for NHTEs, value-added tax (VAT) reform, and CIT collection reform, which may also affect firms' innovation incentives. We take into consideration of them in the robustness part.

¹⁴ See "Provisional Regulations of the People's Republic of China on Corporate Income Tax", published by the State Council in 1993 (No.137).

¹⁵ See "Financial Taxation Issues About Promoting Enterprise Technological progress", published by the Ministry of Finance and the National Tax Bureau in 1996 (No.41).

¹⁶ See "Expansion of Enterprise Scope of Application of the Additional Deduction Tax Policy of Enterprises' Technology R&D Expenses" published by the Ministry of Finance and the National Tax Bureau in 2003 (No.244).

¹⁷ In terms of deduction level, China's policy is comparable to that of the UK, which provides a 150% total deduction of R&D expenses for SMEs, but much higher than that of Norway, which allows a deduction of 20% of the R&D expenses.

2.2. Data

We use the Above-Scale Industrial Firms Panel (ASIFP) dataset maintained by the National Bureau of Statistics (NBS) of China to estimate the (dis)connection between firms' R&D and productivity. The dataset covers all SOEs and non-SOEs with annual sales above 5 million RMB in mining, manufacturing, and utilities industries. It is the most comprehensive firm-level panel data in China. The number of firms in this database varies from approximately 140,000 in 1998 to over 336,000 in 2007. The firms are from all 31 provinces and directly administrated municipalities in China. The ASIFP contains detailed information of each firm, including its official name, age, industrial affiliation, location, and ownership. It also contains a set of firm-level operation and performance items, such as employment, capital, intermediate inputs, new product sales, and R&D investment. We clean the data by removing firms that are not compatible with the basic rules of Generally Accepted Accounting Principles (Cai and Liu, 2009), firms with fewer than eight workers because those firms are subject to a different legal regime (Brandt et al., 2012), and firms with wrong establishment years.

We further curtail the ASIFP panel data for reasons below. First, our sample starts from 2001, the earliest year of available firm-level R&D expenses, to 2007, the year before the financial crisis. Second, we include only manufacturing firms. Third, we exclude all foreign-invested enterprises (FIEs) because (i) they are subject to different CIT regimes from the domestic firms and (ii) their R&D decisions might be influenced by their parent firms abroad. Fourth, because the firm-level R&D investment data are not reported in 2004, we remove all the observations in 2004. Fifth, we remove firms that changed their ownership during our sample period since the CIT incentive program eliminates its discriminatory nature, which is based on the pre-determined registered type, in 2003. Sixth, we exclude firms that did not exist before 2003 or after 2003, the CIT reform year. We finally obtained a firm-level panel covering 2001–2007 around 80 thousand firms and 346 thousand observations across 478 four-digit manufacturing industries.¹⁸

The summary statistics are provided in Table 1. For all observations (Panel A), 14.7 % of firm observations have R&D investment. On average, firms have 483.8 thousand RMB R&D expenditures per year, and the R&D to sales ratio is 0.18 %. The average employment per firm is 295 workers and TFP is 2.549.

For observations with positive R&D investments (Panel B), on average, firms have 3281.2 thousand RMB R&D expenditures per year, and the R&D to sales ratio is 1.25 %. The average employment per firm is 704 workers and TFP is 3.1889. All these numbers are larger than the corresponding ones in the upper panel.

3. The (Dis)Connection between R&D and productivity

In this section, we first establish the fact that the effect of R&D on productivity in China during 2001–2007 is quantitatively small, by strictly applying Aw et al. (2011)'s method of estimating the elasticity of productivity on R&D. The results indicate a disconnection between R&D and productivity at the firm level in China, despite their close co-movements at the macro level.

3.1. Empirical model

In an open economy, firms make their R&D investment decisions jointly with export decisions. Aw et al. (2011) provide a rigorous structural model in which a firm makes its R&D and export decisions in every period, anticipating their impacts on future productivity. This method is intended to be generally applicable in estimating productivity while considering endogenous R&D and exporting. For example, several studies such as Aw et al. (2007) adopt this approach in their studies on Taiwanese electronics industry. König et al. (2022) also use a similar setting to compare productivity between firms in Taiwan and Mainland China. We follow Aw et al. (2011) to estimate the productivity and then explore the real effect of R&D investment on productivity.¹⁹ Specifically, we estimate the effects of Chinese firms' R&D investments on productivity using Equation (8) in Aw et al. (2011) as follows:

$$\omega_{it} = \alpha_0 + \alpha_1 \omega_{it-1} + \alpha_2 (\omega_{it-1})^2 + \alpha_3 (\omega_{it-1})^3 + \alpha_4 d_{it-1} + \alpha_5 e_{it-1} + \alpha_6 d_{it-1} e_{it-1} + \xi_{it},$$

where ω_{it} is firm i 's productivity in year t , d_{it-1} is firm i 's R&D investments in year $t-1$, e_{it-1} is firm i 's export participation in year $t-1$, and ξ_{it} captures firm i 's stochastic nature of productivity improvement in year t .²⁰ The evolution of firm-level productivity is assumed to follow a Markov process, which is affected by the firm's R&D investments, export participation, and a random shock (zero mean and variance). In this equation, the coefficient α_4 is our focus of interest because it captures the direct effect of R&D investments on firm-level productivity. A positive α_4 implies that the firm's R&D investments increase firm-level productivity, while a negative α_4 suggests the opposite effect. In addition, α_5 captures the effect of past exporting activity directly while α_6 captures the interaction effect of both activities.

3.2. Findings

Column (1) in Table 2 reports the estimates of α_4 using the full sample. The result indicates that R&D investments increase firm-level productivity, but the effect is very small: a 1 % increase in R&D investments leads to a 0.00000536 % increase in firm-level

¹⁸ We describe our data processing in Appendix A1.

¹⁹ We describe in details the process of estimating productivity in Appendix A2.

²⁰ In their main analysis, Aw et al. (2011) use d_{it-1} as an R&D dummy. However, they also use it as R&D expenditure in column (2) in Table 2.

Table 1
Summary statistics.

Variables	(1) Mean of all obs.	(2) Mean of control group	(3) Mean of treatment group	(4) St. Dev.	(5) Min	(6) Max	(7) N
Panel A. Full sample							
R&D expenditures	483.7702	397.2733	514.2041	23,070.04	0	7,142,498	340,297
R&D dummy	.1474389	.1223891	.1562526	.3545434	0	1	340,297
R&D to sales ratio	.0018409	.001499	.0019611	.0150778	0	3.09561	340,171
Labor	294.7586	351.2396	274.5106	1199.137	10	147,722	345,891
TFP	2.549035	2.316393	2.631274	1.171516	−7.985	8.737	324,741
Capital to labor ratio	57.16288	47.74754	60.51371	248.9945	.00044	107,142.9	343,168
Foreign capital share	.0027641	.0023663	.0029057	.0402576	0	1	343,168
Export	10,143.42	6351.863	11,502.66	163,214.2	0	5.77e+07	345,891
Age	13.57867	23.61999	9.980071	12.90961	1	101	345,842
Panel B. Sample with positive R&D							
R&D expenditures	3281.158	3245.987	3290.851	60,005.8	1	7,142,498	50,173
R&D participation rate	1	1	1	0	1	1	50,173
R&D to sales ratio	.012484	.0122423	.0125506	.037535	7.23e-07	3.09561	50,161
Labor	704.361	934.6575	630.4665	2697.801	10	147,722	55,767
TFP	3.18886	2.992068	3.244964	1.370342	−7.985	8.737	48,191
Capital to labor ratio	86.4627	68.19011	92.30578	493.5024	.0014	107,142.9	55,540
Foreign capital share	.0040955	.0028391	.0044973	.0435543	0	1	55,540
Export	32,281.78	23,333.25	35,153.07	384,822.1	0	5.77e+07	55,767
Age	16.74221	30.02884	12.48654	15.98507	1	101	55,735

Note: The unit of R&D expenditure (investment) is 1000 RMB. The R&D participation rate is the percentage of firms that have positive R&D. Labor stands for firm employment. TFP represents total factor productivity. Foreign capital share stands for the percentage of firm capital held by foreign investors.

productivity, which represents the average effect of all industries.

A large variation can be observed across industries when we run the regression by industries. For instance, the largest estimate of α_4 is 0.002, which is obtained from the beverage manufacturing industry, whereas the estimates are insignificant in certain industries. α_5 measures the effect of lagged exporting activity on current productivity, which is 0.000581. If firms with exporting experience increase 1 % in previous R&D investments, their productivity will be 0.0000149 % higher. To make sense of the estimates obtained, we compare the estimated α_4 in the electronic industry in the sample, shown in column (2), with that for Taiwan obtained by Aw et al. (2011). The estimate is 0.0001373 from our sample and is 0.0067 for Taiwan as reported in column (2) in Table 2 by Aw et al. (2011), the former being only 1/50 of the latter.

The above results indicate that there is a disconnection between R&D and productivity in China. This is puzzling because we observe a coherent co-movement between R&D and productivity growth at the macro level. This disconnection deserves explanations and warrants important implications for R&D policies. In the next section, we offer one explanation: policy-induced R&D investment is less effective in enhancing firm-level productivity.

4. The 2003 R&D reform and firms' R&D investments

The Chinese government plays a crucial role in influencing firm-level innovation (Ding and Li, 2015; Zilibotti, 2017; Chen et al., 2021; König et al., 2022). In the 2001–2007 period, China introduced a series of fiscal policies to incentivize firm-level innovation activities. We conjecture that firms' R&D decisions are likely to be distorted by these policies and thus cause the inefficiency of R&D investments. In this section, we make use of an R&D-related reform in one of the most notable and pervasive fiscal policies, i.e., the comprehensive CIT incentive program, to evaluate how the R&D tax policy affects the firm-level R&D investments, and how and why the policy-induced R&D leads to the low efficiency in promoting productivity.

4.1. Empirical specification

We evaluate the causal effects of the tax policy on firms' R&D investment. In particular, we make use of the 2003 CIT deduction reform in China and corresponding firm-level data during 2001–2007 to conduct this analysis. One unique feature of the 2003-R&D reform is that some firms (mainly private firms) were eligible for the additional tax reduction policy since 2003, while others (mainly SOEs and COEs) enjoyed the additional reduction in the entire sample period. The variations across firm types and years allow us to employ a DID method to conduct the analysis.

Specifically, the control group consists of firms that have benefited from the tax credit since 1996, which includes SOEs and COEs as discussed in the previous section. The treatment group is the remaining domestic firms, which are mainly private firms. They started to benefit from the tax credit in 2003.

Inspired by Bøler et al. (2015), we introduce the following Ordinary Least Squares (OLS) specification for our baseline DID estimation:

Table 2
Productivity effects of R&D investments.

Variable	(1) Full sample All firms (2001–2007) <i>Productivity</i>	(2) Subsample Electronics (2001–2007) <i>Productivity</i>
α_4	0.0000536*** (0.00000637)	0.0001373 (0.0002274)
α_5	0.0000581*** (0.00000282)	0.0063825*** (0.0014978)
α_6	0.0000373*** (0.00000101)	0.0002489 (0.000345)
Observations	231,002	6,810

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses are clustered at the industry-ownership level.

$$\ln(R\&D_{it}) = \beta_0 + \beta_1 D_i * Post03_t + \gamma X_{it-1} + \lambda_i + \lambda_{jt} + \epsilon_i, \quad (1)$$

where the dependent variable $\ln(R\&D_{it})$ measures the innovation activity of firm i in year t . We adopt the transformed measure, $\ln(R\&D_{it}) = \ln(R\&D_{it} + \sqrt{R\&D_{it}^2 + 1})$, where $R\&D_{it}$ is the R&D expenses, as our dependent variable, which solves the problem of too many zero-R&D observations. D_i is the treatment group dummy, which equals 1 if firm i is in the treatment group, and 0 otherwise. $Post03_t$ is the post-reform period dummy. The value equals 1 for years after 2003. X_{it-1} is a vector of firm i 's time-varying characteristics in year $t-1$, which might affect a firm's R&D decisions, including firm size, capital-labor ratio, foreign capital share, productivity,²¹ export value, age, age squared, and legal registration type.²² We include firm fixed effect λ_i to control for all time-invariant firm characteristics. During our sample period, other contemporaneous reforms, such as China's accession to World Trade Organization (WTO) in 2001 and the FDI entry deregulation in 2002, also occurred. The intensity of these reforms varies across industries and years (Lu et al., 2017; Liu et al., 2019). To identify these confounding effects and other industry-specific trends, we include the industry-year fixed effect, λ_{jt} , where industry j is defined at the four-digit Chinese Industrial Classification (CIC) level. ϵ_i is the robust standard error clustered at the industry-ownership level.

4.2. Treatment and control groups

Our analysis is based on two data sources: the policy document data and firm-level balance sheet data. We have described the policy document data in Section 2.1 and the firm-level balance sheet data, i.e., ASIFP, in Section 2.2.

We define our treatment group by adhering strictly to the corresponding definitions in the CIT deduction reform in 2003. The government documents provide details of the preferential tax treatments for R&D investment. They include “Financial Taxation Issues about Promoting Enterprise Technological Progress” published by the Ministry of Finance and the National Tax Bureau in 1996, “Management Measurements for Pre-tax Deduction of Enterprises' Technology R&D Expenses” published by the Torch Center of the Chinese Ministry of Science and Technology, the Ministry of Finance, and the National Tax Bureau in 1999, “Expansion of Enterprise Scope of Application of the Additional Deduction Tax Policy of Enterprises' Technology R&D Expenses” published by the Ministry of Finance and the National Tax Bureau in 2003, and “Preferential CIT Policies on Enterprise Technology Innovation” and “Relevant Questions about Clarifications on the Implementation of CIT” published by the Ministry of Finance and the National Tax Bureau in 2006. A key to classifying firms into the control and treatment groups is ownership type, which is identified in the ASIFP data by the firms' legal registration type. As for joint-stock and joint-venture enterprises, we put them in the control group if a firm has at least 25 % capital share from state or collective ownership.²³

The summary statistics for the sample of the control group and that of the treated group are reported in columns (2) and (3) in Table 1, respectively.²⁴ Panel A includes all observations, with or without R&D investment. Among all observations, 73.6 % of the observations belong to the treatment group and the rest belong to the control group. In the treatment group, the average R&D expenditure per year is 514.2 thousand RMB per year, 15.6 % of the observations have R&D investment, and the average R&D to sales ratio is 0.2 %. In the control group, firms on average have 397.3 thousand RMB R&D investments per year, 12.2 % of the observations perform R&D activities, and the average R&D to sales ratio is 0.15 %. On average, firms in the control group have a larger size (labors),

²¹ Firm-level productivity is calculated based on Levinsohn and Petrin (2003) at the two-digit CIC industry level and using the information of employment, fixed assets, value added, and intermediate input. We deflate all nominal variables based on the deflators in Brandt et al. (2012).

²² Within the treatment and control ownership groups, around 27% of firms switched their legal registration types during our study period. Hence, we control for firm legal registration type in firm characteristics.

²³ We also consider joint-stock enterprises and joint-venture enterprises with at least 50% capital share from state or collective ownership. The patterns of our empirical results do not change.

²⁴ See Appendix A3 for more information.

smaller export value, smaller R&D expenditure, lower productivity, and smaller capital to labor ratio, compared with those in the treatment group.

Moreover, for observations with positive R&D investments (Panel B), around 25 thousand firms and 56 thousand observations across 471 four-digit manufacturing industries can be found. Similar comparisons between the two groups in various dimensions are the same as those in Panel A.

4.3. Empirical results

We report our baseline results in Table 3. In column (1), the simple correlation (without any controls and fixed effects) between R&D investment and the interaction term $D_i \cdot \text{Post03}_t$ is presented and the estimated coefficient is significantly positive. In column (2), we add industry-year fixed effects to control for industry-specific trends and other industrial policies such as tariff reductions. The estimate of $D_i \cdot \text{Post03}_t$ is statistically significant and positive. In column (3), we further include firm fixed effects to determine the influence of time-invariant firm characteristics and whether the pattern of the result remains unchanged. In column (4), we introduce lagged one-period time-varying firm characteristics that might have effects on firms' R&D decisions, including firm size, capital-labor ratio, foreign capital share, productivity, export value, age, age squared, and registration type. These additional controls do not change the sign and significance of the effect of the 2003 R&D reform on firms' R&D investments. In sum, the findings indicate that the 2003 R&D reform effectively increases the policy-targeted firms' (i.e., private firms) R&D investments.

The effect is also economically significant. Based on estimates in column (4), the 2003 R&D reform induces the treated firms to increase their R&D investment by 6.68 %, which is equivalent to an additional 17,028 RMB R&D investment of each treated firm, or 109,570 RMB of each treated firm with positive R&D investment as compared with R&D investment of treated firms before the reform.²⁵ In addition, the estimated elasticity of R&D with respect to the tax deduction for the treated firms is -0.9147 , which is statistically significant at 1 %. Therefore, a one-unit tax deduction could simulate about 0.91 units of R&D expenditures.²⁶ This finding, i.e., the around-unity negative elasticity, is on par with most existing studies on tax effectiveness, such as Hall and Van Reenen (2000), Bloom et al. (2002), and Becker (2015).

We also examine the effects of the reform on firm-level R&D intensity (R&D investments/sales) and R&D participation. The estimated results are reported in Appendix A5. We find that the reform increased the firm-level R&D intensity but does not have a significant effect on the firm-level R&D participation. The pattern of the results suggests that firms with positive R&D investments before the reform are the driving force in exerting positive effects of the reform. We further estimate the policy effect on firm-level R&D investments using a subsample of the firms with positive pre-2003 R&D in Appendix A5. The estimated coefficient of the DID term is 0.1300, twice as much as the baseline coefficient in column (4) in Table 3, indicating a larger policy effect.

4.4. Robustness checks

4.4.1. Flexible estimation

In our baseline results, we estimate the average treatment effect of the 2003 R&D reform, that is, the difference between the treatment and control groups in their average differences in R&D expenses between the pre- and post-reform periods. Now, we check the differences with a more flexible specification of the empirical model. Specifically, we conduct the flexible estimation by replacing $D_i \cdot \text{Post03}_t$ with $\sum_t D_i \cdot \lambda_t$ in the model (1), with λ_t indicating years from 2002 to 2007 and 2001 as the omitted reference year. This check helps answer two questions: whether the two groups have comparable trends before the policy and whether the divergence of trends in difference emerges after the policy.

The estimated coefficients of $D_i \cdot \lambda_t$ by year are shown in column (1) in Table 4 and drawn in Fig. 2. The results show that the estimated coefficient of $D_i \cdot \lambda_{2002}$ is insignificant, indicating that the R&D investment between the treatment and control groups has no significant difference in the pre-reform period as compared with the difference between the two groups in 2001. In addition, the estimated coefficients become significantly positive with increasing magnitudes in the post-reform period since 2003. The estimated coefficient of $D_i \cdot \lambda_{2003}$ is positive but only at a 10 % significance level, which could be because the policy was announced at the end of 2003 and most firms could not respond accordingly and effectively within such a short time. We will elaborate on this point in more detail in Section 4. The coefficients after 2003 are all positive and significant. Overall, these results support our DID specification.

4.4.2. SOE specific trends

We have identified the policy effect by checking the differences between the treatment and control groups in R&D investment between the pre- and post-reform periods. However, our results could be contaminated if the treatment and control groups have different trends in conducting R&D. This is a legitimate concern, especially in our study where our control group is composed mainly of SOEs and COEs and our treatment group with private firms. In terms of the incentive scheme, management, and operation, SOEs are different from the other two types. We may conjecture that SOEs may have a different trend in conducting R&D investment, and if that is the case, our estimates could be potentially contaminated. To address the concern that our results are driven by ownership-specific

²⁵ $17,028 \text{ RMB} = 6.68\% \times 254,905.4 \text{ RMB}$ (the average R&D expenses of private firms in our full sample before the reform), and $109,570 \text{ RMB} = 6.68\% \times 1,640,266 \text{ RMB}$ (the average R&D expenses of private firms with positive R&D investment before the reform).

²⁶ See Appendix A4 for the detailed examination.

Table 3
Baseline results.

Variable	(1) <i>ln(R&D)</i>	(2) <i>ln(R&D)</i>	(3) <i>ln(R&D)</i>	(4) <i>ln(R&D)</i>
$D_i \text{ *Post03}_t$	0.2575*** (0.0445)	0.3188*** (0.0303)	0.1650*** (0.0215)	0.0668*** (0.0231)
$\ln(\text{labor})_{it-1}$				0.2277*** (0.0181)
$\ln(\text{kl})_{it-1}$				0.0403*** (0.0080)
$\text{foreignShare}_{it-1}$				−0.0442 (0.1770)
tfp_{it-1}				0.0885*** (0.0092)
$\ln(\text{age})_{it-1}$				−0.0619* (0.0369)
$\ln(\text{age}^2)_{it-1}$				0.0132 (0.0095)
$\ln(\text{export})_{it-1}$				0.0135*** (0.0027)
RegType_{it-1}				Y
Industry-Year FE	N	Y	Y	Y
Firm FE	N	N	Y	Y
Observations	340,297	340,276	338,763	263,272
R-Square	0.003	0.116	0.674	0.702

Note: Registration type_{it-1} includes a set of dummy variables indicating the firm's registered type in year t-1. We do not report the coefficients to save space. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses are clustered at the industry-ownership level.

Table 4
Validity of the model specification.

Variable	Parallel trend tests (1) <i>ln(R&D)</i>	Ownership time trends (2) <i>ln(R&D)</i>	COE only in control (3) <i>ln(R&D)</i>	COE only in control (4) <i>ln(R&D)</i>	PSM groups (5) <i>ln(R&D)</i>	PSM groups (6) <i>ln(R&D)</i>	WTO shock (WTO dummy) (7) <i>ln(R&D)</i>	WTO shock (Input tariff) (8) <i>ln(R&D)</i>
$D_i \text{ *Post03}_t$		0.1225*** (0.0241)		0.1487*** (0.0267)		0.0570** (0.0247)	0.0599***	0.0668***
$D_i^* \lambda_{2002}$	0.0299 (0.0198)		0.0041 (0.0238)		0.0296 (0.0197)		(0.0226)	(0.0238)
$D_i^* \lambda_{2003}$	0.0382** (0.0193)		0.0058 (0.0250)		0.0298 (0.0191)			
$D_i^* \lambda_{2005}$	0.0691** (0.0301)		0.0952*** (0.0330)		0.0527* (0.0317)			
$D_i^* \lambda_{2006}$	0.0855** (0.0332)		0.1554*** (0.0403)		0.0666* (0.0345)			
$D_i^* \lambda_{2007}$	0.1466*** (0.0346)		0.2349*** (0.0407)		0.1394*** (0.0386)			
Firm controls	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	263,272	263,272	155,431	155,431	214,442	214,442	263,272	263,272
R-Square	0.702	0.702	0.635	0.634	0.703	0.703	0.702	0.702

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses are clustered at the industry-ownership level.

trends instead of the 2003 R&D reform, we further include the SOE specific trend into our regression.

Specifically, we add an interaction term of the SOE dummy with year t to control for SOE ownership-year trends. This specification allows SOEs to have their linear time-trend of R&D investment. Following [Chen et al. \(2019\)](#), we include state-owned sole enterprises (registration type 110), state-owned joint venture enterprises (registration type 141), and state-owned and collective joint venture enterprises (registration type 143) in our SOE group. The value of the SOE dummy is 1 if a firm belongs to this group, and 0 otherwise. Our data sample contains 11,036 SOEs across 466 industries. Column (2) shows that our finding remains robust after controlling for the SOE-specific trend.

4.4.3. COE as the control group

The difference between SOEs and private firms in their R&D activities is the major concern on the comparability between the treatment and control groups. According to [Du and Girma \(2009\)](#) and [Zhou et al. \(2017\)](#), the management and operation of COEs are similar to private firms. Therefore, we expect that COEs and private firms are also similar in their R&D activities. Accordingly, we

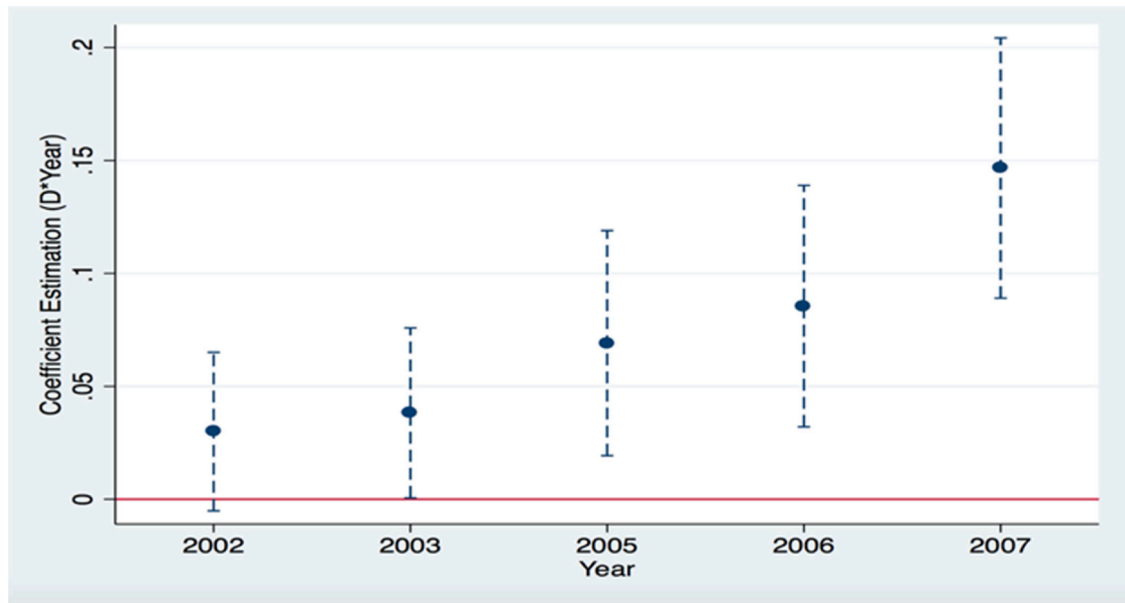


Fig. 2. Coefficients of the estimation.

Note: Dash lines are confidence intervals at the 90 % significance level.

exclude SOEs and just keep COEs in our control group to re-run our baseline regression. If our baseline results are driven by SOE-specific trends, we expect to find no significant difference in R&D investment between COEs and private firms after the reform. If we find significant differences between them, then our baseline finding is not driven by the potential different trends in R&D investment of SOEs.

Using COEs as our control group, we first run a flexible estimation. The results in column (3) in Table 4 show that COEs and private firms do have comparable pre-trends. The small coefficients of the pre-reform periods also suggest that these two groups of firms have similar trends in R&D behavior. In addition, the average treatment effect of the reform, as shown in column (5), remains significantly positive, confirming our baseline finding.

4.4.4. Matched control group

Besides ownership-specific trends, SOEs and COEs may also differ from private firms in some other dimensions. To address the concern, we use the PSM method to select a subsample of untreated firms with similar characteristics to private firms. In particular, we utilize the observations in the initial year 2001 and select the most comparable non-private firms based on firm-level labor, capital-labor ratio, foreign capital share, total sale and total asset to construct our control group.²⁷ In this matched subsample, the number of firms is 55,906 with 66.1 % of them belonging to the treatment group. We first find that the pre-trends between this newly matched control group and the treatment group are comparable, as shown in column (5) in Table 4. Then, we re-run model (1) with this new subsample and report the results in column (6). The finding is robust.

4.4.5. WTO shock

China joined the WTO at the end of 2001. This reform has dramatically reduced the import tariffs and uncertainty of trade policies across industries and had significant effects on Chinese firms' incentives for R&D investment (Liu and Qiu, 2016; Liu et al., 2021). Because these changes occurred at the industry level, their effects have been mostly accounted for by our industry-year fixed effects. However, a concern that the treatment and control groups may react to the same WTO shock differently remains. To address this issue, we add an interaction term of the treatment group dummy and WTO dummy, $D_i \cdot WTO_t$, where WTO_t equals 1 for all years after 2001 and 0 otherwise, to control for the differential effects across our treatment and control groups. Results in column (7) in Table 4 confirm that our estimation is robust to the shocks from China's WTO entry.

We further use China's annual import tariff at the industry-year level during our sample period to capture the WTO effects because the WTO entry directly drove down the tariff levels. Specifically, we add an interaction term of the treatment group dummy and industrial tariff, $D_i \cdot Tariff_{it}$, where $Tariff_{it}$ represents annual input tariff at the four-digit industry level, in our baseline regression, to control for the different WTO effects across the treatment and control groups. The results are presented in column (8). We find that the coefficients of our DID term are still significantly positive, indicating that the patterns of the results do not change.

²⁷ The matching regression results and the balance property test results are presented in Appendix A6.

4.4.6. Other related tax policies

During our sample period, the 2003 R&D reform was the only CIT policy reform that directly targeted private firms (our treatment group). However, several CIT-related policies, though not targeting private firms, were also in effect or introduced, which might affect firms' R&D incentives. In this subsection, we take those policies into account to determine whether our findings are robust to them. Specifically, we explore three influential CIT policies.

First, we consider the InnoCom Program, which is specially designed for NHTEs and has been in effect since 1991.²⁸ If a firm is qualified for the InnoCom Program, it can enjoy a preferential CIT rate of 15 % while other unqualified firms pay CIT at the standard rate of 33 % before 2008.

During our sample period, a key requirement of the InnoCom Program is that a firm's R&D intensity (i.e., R&D to sales ratio) should be no less than 5 % if the firm is located inside a high-tech zone, and no less than 4 % if the firm is located outside a high-tech zone. In our sample, approximately 3 % of the firm-level observations have R&D intensity equal to or greater than 4 %. Hence, we remove those observations from our sample and re-run the regression based on the specification (1). The result in column (1) in Table 5 shows that our findings are robust and not driven by the InnoCom Program.

Second, in 2004, three provinces in northern China, namely, Heilongjiang, Jilin, and Liaoning, began a pilot reform on VAT. The reform allowed firms in these three provinces to deduct the fixed assets investment from their VAT base.²⁹ Given that R&D investment is associated with fixed assets investment, we can expect that such policy might also encourage R&D investment. Although this VAT reform does not discriminate between our treatment and control groups, we cannot rule out the possibility that the two groups may respond differently. We remove all observations from these three provinces, which account for around 5 % of all observations in our sample, to mitigate the potential concern caused by this 2004 VAT pilot reform. We re-run the model (1) using this censored sample and present the empirical results in column (2). We obtain similar significantly positive effects as in the baseline results. Therefore, the 2004 VAT pilot reform does not bias our baseline results.

Third, as mentioned in Cai et al. (2018), the effective CIT rates are statistically different between Chinese firms that pay CIT to local tax bureaus and those that pay to state tax bureaus after 2002. To tease out the potential confounding effect from this reform, we remove firms eligible for the CIT collection reform and able to pay directly to state tax bureaus, which accounts for approximately 2 % of the total observations. We re-run the regression based on the model (1) and report the results in column (3). The results are consistent with our baseline findings.

Our baseline findings may also be driven by other CIT policies. Hence, we conduct a further placebo test to address these concerns. Specifically, we keep firms that need to pay CIT but are not eligible for the additional deduction of R&D expenses based on firms' annual R&D growth rate.³⁰ Therefore, these firms are exposed to other CIT policies, if any, while not exposed to the R&D deduction policy. If we find a non-positive effect in this placebo test, we can confidently state that our baseline estimate does capture the effect of the R&D deduction policy. Results in column (4) show that no significant effect can be observed, thereby confirming that our baseline findings are not driven by other CIT policies.

4.4.7. Alternative estimation method

In our baseline empirical analysis, we use the OLS method to investigate the impact of the 2003 reform on firm R&D investments. To further address the concern of the zero-R&D investments, we employ the Poisson pseudo-maximum likelihood (PPML) estimation described by Silva and Tenreiro (2006) to explore the effect. We present the results in column (5) in Table 5. The positive and significant coefficient of the DID term suggests the consistency with our main finding.

4.4.8. R&D expenditure in 2004

Our original data do not have R&D expenditure in 2004, which might cause concerns in estimation. Thus, we infer R&D expenditure in 2004 by calculating the average value of the R&D expenditure in 2003 and 2005 and then re-run the model (1) with the inferred 2004 data included. Results in column (6) in Table 5 show that our findings are robust when including the inferred R&D expenditure in 2004.

4.4.9. Full sample

In the baseline analysis, our sample retains firms that remained operation in at least one year before 2003 and at least one year after 2003, to ensure that they all experienced the impact of the 2003 reform. Several existing studies, such as Brandt et al. (2012), show that firms' enter and exit is one of main drivers of China's growth during 1998–2007. In order not to ignore those firms, we put them back to our sample to have a robustness check. We re-run the model (1) with this larger sample and report results in column (7) in Table 5. The result shows an even larger effect of the reform on the firm-level R&D.

4.4.10. Potential serial correlation of R&D

One may be concerned that R&D investment usually is a lumpy and dynamic decision, and previous R&D may have critical effects on current R&D expenditure. That is, R&D investments over time may be serially correlated. We include the firm's previous R&D in our baseline regression to mitigate this concern. We show the results in column (8) in Table 5, which suggests that our main finding is not

²⁸ See Chen et al. (2021) for a detailed discussion.

²⁹ Howell (2016) explores the effects of this reform on firm R&D in China.

³⁰ A no-less-than 10% R&D growth in two consecutive years is a requirement for firms to enjoy the preferential tax deduction.

Table 5
Contemporary policy changes and other samples.

Variable	InnoCom Program (1) <i>ln(R&D)</i>	2004 VAT reform (2) <i>ln(R&D)</i>	CIT collection reform (3) <i>ln(R&D)</i>	Unqualified CIT payers (4) <i>ln(R&D)</i>	PPML method (5) <i>ln(R&D)</i>	Inferred 2004 R&D (6) <i>ln(R&D)</i>	Full sample (7) <i>ln(R&D)</i>	Serial correlation (8) <i>ln(R&D)</i>
$D_i \cdot \text{Post03}_t$	0.0505** (0.0209)	0.0600*** (0.0235)	0.0631*** (0.0230)	−0.2335 (0.2265)	0.0537** (0.0256)	0.0988*** (0.0182)	0.1253*** (0.0238)	0.0607*** (0.0235)
Firm controls	Y	Y	Y	Y	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y	Y	Y	Y	Y
Observations	259,916	249,217	258,920	11,573	78,906	311,706	332,983	231,595
R-Square	0.687	0.701	0.702	0.734		0.722	0.739	0.721

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses are clustered at the industry-ownership level.

driven by the potential serial correlation of R&D.

4.4.11. Potential treatment heterogeneity

De Chaisemartin and d'Haultfoeuille (2020) demonstrate that if serious heterogeneous treatment effects exist across groups and periods, the two-way fixed effects model could suffer from the negative weights problem that the estimate would be biased. We follow the method proposed by De Chaisemartin and d'Haultfoeuille (2020) to check the share of negative weights in our baseline estimation and estimate the effect of the 2003 reform on firm-level R&D. We find that only 0.46 % of our divisions of group and period have negative weights. The estimated coefficient is still significant and positive at 0.0773, suggesting that the 2003 reform increases the treated firms' R&D by 7.73 %. Therefore, our baseline findings are robust to this estimation concern.

4.5. Heterogeneous effects of the 2003-R&D reform

We have verified the average effects of the 2003 R&D reform. However, firms with different characteristics such as R&D capabilities may respond to the reform differently. Hence, we analyze the heterogeneous effects to gain more understanding of the policy effects.

4.5.1. Firms with different profits

We investigate the heterogeneous effects of the 2003 R&D reform on R&D investments of firms with different profits. Given that CIT payable is calculated based on profits, firms with higher profits may have stronger incentives to apply for the CIT deduction of R&D expenses. Specifically, we classify firms into low-and high-profit groups in each industry according to whether the firm's initial profit level is lower or higher than the median of its industry, with the initial profit defined as the average value of firm-level profits in 2001 and 2002. We use the initial profit level before 2003 to avoid the endogeneity that profits after 2003 might be affected by the 2003 R&D reform. We present the results using the subsamples of the low-profit and high-profit groups in columns (1) and (2) in Table 6, respectively. The results indicate that the average effect of the 2003 R&D reform is driven mainly by firms in the high-profit group, which confirms the importance of high-profit firms. These results confirm our expectation that firms with larger profits have higher incentives to increase their R&D investment for the CIT deduction.

4.5.2. Firms with different productivity levels

Firms' innovation abilities may be bounded by their levels of productivity. For example, firms with frontier technologies usually

Table 6
Heterogeneous effects.

Variable	(1) Subsample low-profit <i>ln(R&D)</i>	(2) Subsample high-profit <i>ln(R&D)</i>	(3) Subsample low-tpf <i>ln(R&D)</i>	(4) Subsample high-tpf <i>ln(R&D)</i>
$D_i \cdot \text{Post03}_t$	−0.0047 (0.0243)	0.1024*** (0.0375)	−0.0097 (0.0252)	0.0842** (0.0389)
Firm controls	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y
Observations	116,508	136,190	94,717	121,806
R-Square	0.685	0.724	0.638	0.739

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses are clustered at the industry-ownership level. Other interaction terms are included but not reported to save space.

invest more in innovation activities to compete with other leading firms whereas firms far from the frontier technologies tend to spend less on R&D projects and prefer to imitate rather than innovate (König et al., 2022). Therefore, firms with lower productivity may be less sensitive to the 2003 R&D reform compared with the frontier firms. We explore this heterogeneous effect by dividing firms into low- and high-productivity groups in each industry according to the industry's initial median productivity, analogously to the case of profit classification. The subsample estimation results in columns (3) and (4) in Table 6 show that the effect is driven mainly by high-productivity firms. Therefore, highly productive firms benefit more from the 2003 R&D reform.

5. The 2003 R&D reform and the effect of R&D on firms' productivity

Thus far, we have established that the 2003 R&D reform increases firms' R&D expenses. However, do these R&D investments generate real effects on firms' performances? We explore this question in this section.

We first estimate the effects of R&D investments on firm-level productivity based on the subsample of all treated firms, using Equation (8) in Aw et al. (2011). The result in column (1) in Table 7 indicates that a 1 % increase in R&D investments increases firm-level productivity by 0.00000333 % for the treated firms.³¹ Recall that our baseline finding shows that the 2003 R&D reform raises private firms' R&D investments by 6.68 %, the reform thus induces a 0.00002224 % ($= 6.68 \times 0.00000333$) increase in firm-level productivity by the treated firms. The overall real effect of the reform is small. Moreover, we explore alternative specifications with a different set of controls, including firms' past importing activity, industry-year fixed effects, and city-year fixed effects. We also employ the two-stage IV regression method and instrument the past R&D in the regression model to investigate the real effects of R&D investments on firms' productivity. We report these results in Appendix A8. The estimates are all significantly positive, consistent with our main finding.

Since the positive impacts of the 2003 reform on R&D are driven by the treated firms with positive R&D (excluding firms that do not conduct R&D), we also present the effects of R&D on firms' productivity using a subsample of the treated firms with positive R&D in column (2). The elasticity for the treated firms with positive R&D is 0.00000896, two times larger than that for all treated firms.

We further separately estimate the R&D-productivity elasticities for our treatment and control firms in both the pre- and post-reforms. We report estimates using firms with positive R&D in the four subsamples in columns (3)–(6) since the policy effects were mainly driven by firms with positive R&D investments. We find that the elasticity of R&D on productivity for the treated firms has drastically decreased from the pre-reform to post-reform periods, but not for the control firms. The elasticity for treated firms in the post-reform period is only 30 % of that in the pre-reform period. In contrast, the elasticity for the control firms almost stays the same in the pre-reform period as that in the post-reform. The fact that R&D investments become less efficient after the 2003 R&D reform for the treated firms suggests that the policy-induced R&D investments are not as effective as the spontaneous R&D in promoting productivity.

The above findings suggest that the aforementioned two explanations, i.e., technology frontier and misallocation, cannot fully explain the low efficiency of R&D investment on productivity. It is true that had China had technology closer to the world frontier, creating new ideas might have been more efficient and contributed more to firms' productivity. However, this argument cannot reconcile our findings that the R&D of the treated firms become less efficient in promoting productivity in the post-reform period when China's technologies are closer to the frontier. As for the misallocation argument, the 2003 R&D reform transfers a previous discriminatory policy towards a non-discriminatory one, which should help reduce the misallocation of R&D activities across the treated and control firms. If we believe that the *ex-ante* allocation is inefficient, this reform should on average increase the R&D efficiency of the treated firms.

In the next section, we add another explanation: relabeling. Chen et al. (2021) show that under the InnoCom program in China from 2008 to 2011, firms had incentives to relabel some non-R&D expenditures as R&D expenditures to qualify for the program and obtain more subsidies. In the next section, we provide evidence on relabeling under the 2003 R&D reform during the period 2001–2007. Because relabeling over reports R&D investments, it hampers R&D efficiency.

6. The 2003 R&D reform and relabeling

6.1. Evidence of relabeling

Several particular features of the 2003 R&D reform enable us to identify relabeling of R&D expenses. The reform was announced on November 27, 2003, nearly the end of the fiscal year in 2003. Although all R&D expenses in 2003 are eligible for the deduction, it is reasonable to believe that it would have been difficult for firms to conduct large-scale real R&D investments in such a short time (around one month) after the announcement. Therefore, if we observe a sudden increase in reported R&D investment in 2003 compared with 2002, we can confidently claim that a relabeling problem exists. We present some evidence and analyses of relabeling in this subsection.

The first is the 10 % criterion. As described in subSection 3.1, a no-less-than 10 % R&D growth in two consecutive years is a requirement for firms in the treatment group to enjoy the preferential tax deduction after the 2003 R&D reform. We suspect that some firms in the treatment group manage to achieve such a growth target. Fig. 3 plots the density distribution of firms by R&D growth rates within the range of 5 %–15 % with firms in the treatment group in the left panel and firms in the control group in the right panel. In

³¹ We also examine the direct effect of the reform on firm-level productivity using the DID method, and find consistent results. Please see Appendix A7 for details.

Table 7
Productivity effects of R&D investments.

	(1) Subsample Treated firms	(2) Subsample Treated firms (Positive R&D)	(3) Subsample Treated firms (Positive R&D) (before 2003)	(4) Subsample Treated firms (Positive R&D) (after 2003)	(5) Subsample Untreated firms (Positive R&D) (before 2003)	(6) Subsample Untreated firms (Positive R&D) (after 2003)
Variable	Productivity	Productivity	Productivity	Productivity	Productivity	Productivity
α_4	0.00000333*** (0.00000708)	0.00000896*** (0.00000204)	0.0000111*** (0.00000302)	0.00000335 (0.00000279)	0.0000181*** (0.0000051)	0.0000123** (0.00000634)
α_5	−0.00000826*** (0.00000305)	0.00000483 (0.0000188)	0.0000207 (0.0000271)	−0.0000263 (0.0000263)	0.0000146 (0.0000514)	0.0000331 (0.0000653)
α_6	0.00000658*** (0.00000110)	0.00000321 (0.00000296)	0.00000141 (0.00000462)	0.00000720* (0.00000396)	−0.00000210 (0.00000859)	−0.00000368 (0.00000956)
Observations	173,253	27,601	11,754	15,847	4,146	3,287

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses.

each panel, we have the distributions of three years, which are 2002, 2003, and 2006.³²

We observe a clear notching of firms with R&D growth rates around 10 % in 2003 and 2006, compared with 2002 in the treatment group in Panel A. However, such a pattern does not exist in the control group in Panel B. While the notching outcome in 2006 is consistent with the incentive responses of firms in their real R&D investments, the notching outcome in 2003 seems not. There are many reasons for the observation that many firms concentrate in the region slightly below the 10 % threshold. For example, firms did not have sufficient time to adjust their R&D growth rates after the announcement of the reform in late 2003. This “puzzled” observation also presents in other studies.³³

The second is the ratio of R&D to administrative expenses. If relabeling happens, how? We check the ratio of R&D to administrative expenses of firms around the 10 % threshold. In China, R&D expenditure is part of the total administrative expenses. Fig. 4 depicts the ratio of R&D to administrative expenses of firms in the treatment group whose R&D growth rates are around the required threshold (from 5 % to 15 %) and the quadratic fitted lines in the pre- and post-reform periods. The figure shows that a discontinuous upward jump exists at the 10 % threshold in 2003 and 2006, but not in 2002.³⁴ Although it is difficult to increase R&D investment, especially in 2003, firms can switch some administrative expenses to R&D expenses, which will result in a sudden increase in the R&D-to-administrative expenses ratio. The observed discontinuous jump is consistent with this story.

Third is the non-R&D administrative expenses. We cannot rule out the possibility that firms manage to increase their R&D-to-administrative expenses ratios by increasing real R&D investment, even in 2003. If this is the case, we should expect to observe an increase in R&D expenses but stable non-R&D administrative expenses. We now show that this is not the case, that is, firms have fewer non-R&D administrative expenses, which suggests the behavior of relabeling some non-R&D administrative expenses as R&D expenses.

Fig. 5 plots the patterns of the ratios of non-R&D administrative expenses to sales of firms in the treatment group in 2002, 2003, and 2006. We focus on firms with R&D growth rates around the threshold from 5 % to 15 %. Panel A plots the ratios of non-R&D administrative expenses to sales in 2002. Panel B plots the ratios in 2003 while Panel C plots the ratios in 2006. An obvious discontinuous drop at the threshold in Panel C, but not A. Panel C indicates that firms that manage to achieve the threshold have lower ratios of non-R&D administrative expenses to sales after the reform. This drop in non-R&D administrative expenses is not because of the reduction in operation and business since firms' sales stay the same around the threshold in these years.³⁵

In sum, we find evidence that relabeling behavior existed in 2003. Similar patterns observed from data in 2003 are also observed from data in 2006. However, we can reasonably believe the relabeling behavior in 2006 is less serious than that in 2003.

6.2. Relabeling and R&D efficiency

In the previous subsection, we have shown that firms with R&D growth rates ranging from 5 % to 15 % are most likely to relabel their R&D investments. It should be clear that relabeling incentives also exist for firms not in the 5 %–15 % R&D-growth range because

³² We do not have the growth rates for 2004 and 2005 because R&D data in 2004 are missing and we do not present the growth rate for 2007 because this year is no longer relevant because of the cancellation of the requirement after 2006.

³³ The increased concentration of firms is not only around the region slightly above 10% (to meet the threshold) but also around the region slightly below 10%. While the below 10% concentration is not consistent with the response to the new policy in theory, it could be explained that in practice, some firms try to achieve the 10% target but fail. This phenomenon is commonly observed, for example, in the study of the InnoCom Program by Chen et al. (2021). Panel A of Fig. 3 in Chen et al. (2021) shows a similar pattern that a larger density of firms with R&D growth rate just below the required threshold for corporate tax deduction than that just above the threshold.

³⁴ We also explore the effects based on Regression Discontinuity Design (RDD) method and the results shown in Appendix A9 are consistent with our findings.

³⁵ We investigate the effects based on RDD method and the results shown in Appendix A9 suggest a similar pattern.

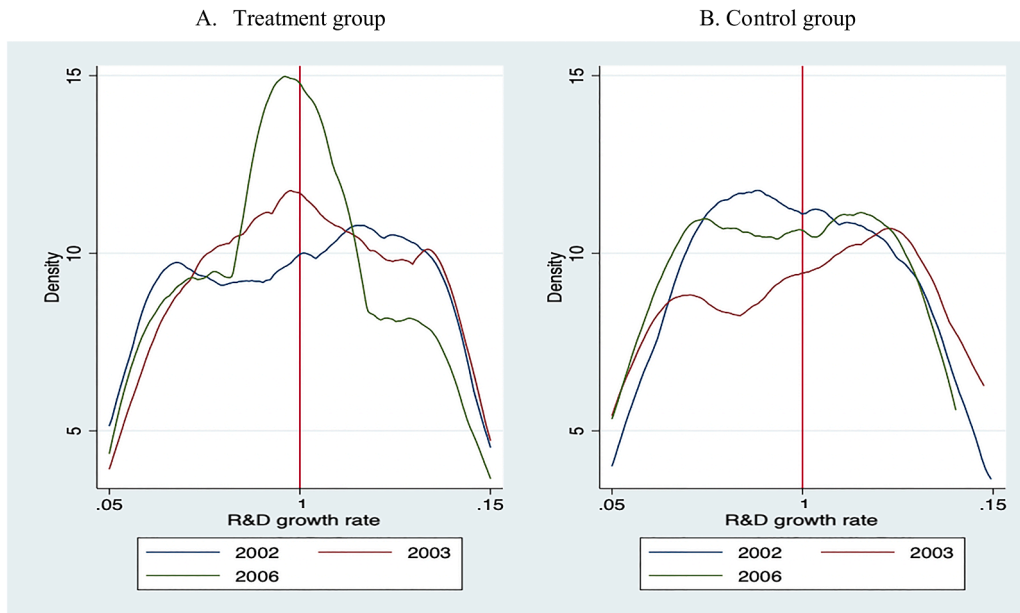


Fig. 3. Distribution of firms by R&D growth rates.

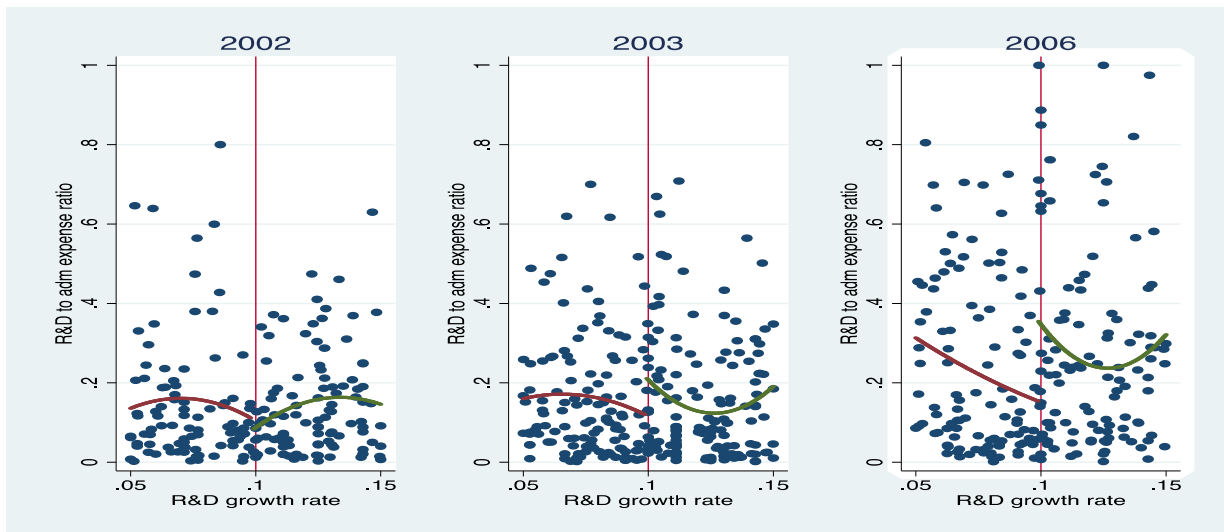


Fig. 4. Ratios of R&D to administrative expenses in the treatment group.

relabeling can still allow them to receive more subsidies under the policy.

We now estimate the R&D efficiency (α_4) based on different subsamples of the treated firms. First, we keep the treated firms with R&D growth rates ranging from 5 % to 15 % for at least one year after the 2003 reform and re-estimate the R&D efficiency (α_4). We expect the R&D efficiency of treated firms in this subset to be limited as a larger proportion of these firms' R&D investments is likely to be fake and unproductive. The result is reported in column (1) in Table 8, which shows that the estimated elasticity is negative and insignificant, confirming that relabeled R&D investments do not enhance productivity.

Second, we estimate the R&D efficiency by deleting the 5 % to 15 % R&D growth firms. By deleting the treated firms with the strongest incentives to misreport, we expect to observe a higher R&D efficiency. This is confirmed by column (2): the estimated elasticity becomes positive and significant and larger than the average elasticity for the treated firms with R&D growth rates ranging from 5 % to 15 %.

Third, we explore heterogeneous effects of R&D investments on productivity for the treated firms with high relabeling incentives. We define a dummy variable *RelabelFirm*, which is equal to 1 if the firm has an R&D growth rate ranging from 5 % to 15 %. We employ the DID method to estimate the heterogeneous effects by adding the relabeling-firm dummy, and an interaction term of firms' past

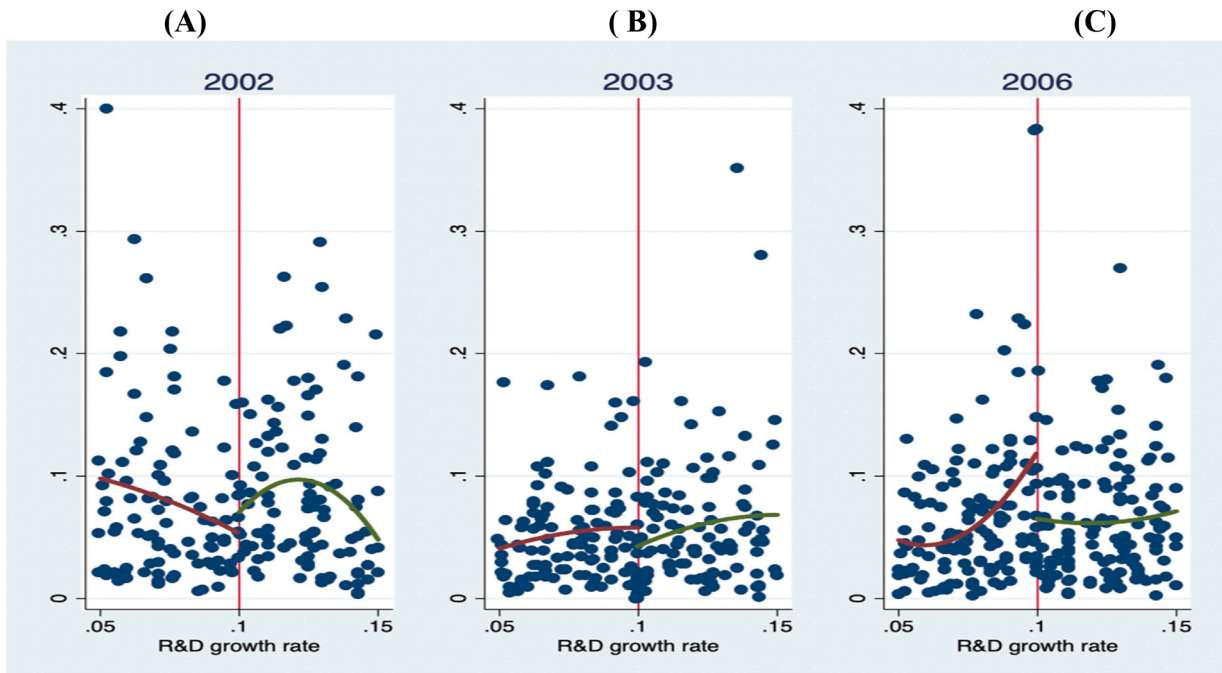


Fig. 5. Ratios of non-R&D administrative expenses to sales.

Table 8
Productivity effects of R&D investments.

Variable	(1) Subsample Treated firms (Relabel firms) <i>Productivity</i>	(2) Subsample Treated firms (Excluding relabel firms) <i>Productivity</i>	(3) Subsample Treated firms <i>Productivity</i>
α_4	−0.00000113 (0.00000292)	0.00000285*** (0.000000749)	0.00000345*** (0.000000804)
α_5	0.0000639** (0.0000279)	−0.00000902*** (0.00000307)	0.00000976*** (0.00000338)
α_6	−0.00000566 (0.00000429)	0.00000698*** (0.00000118)	0.00000608*** (0.00000121)
$d_{it-1} * RelabelFirm_i$			−0.00000506* (0.00000282)
Observations	3,675	169,578	173,253

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors in parentheses.

R&D investments and the dummy to the specification. The results are reported in column (3). The DID estimation results suggest that R&D of the firms with high relabeling incentives is not efficient in improving productivity.

In sum, these results indicate that relabeling partly explains the inefficiency of R&D investments on productivity. Our contribution to this third explanation for low R&D efficiency in China is different from that of [Chen et al. \(2021\)](#). After observing relabeling behavior, [Chen et al. \(2021\)](#) use a structural model to estimate the productivity effect of real R&D by assuming that all fake R&D is unproductive and therefore exclude it from the productivity estimation. They find high efficiency of (real) R&D. Their finding indirectly implies that R&D efficiency would be reduced should the fake R&D be included in the total R&D investments. Our study supplements theirs. We think it is difficult to estimate the percentage of total R&D belonging to relabeling or fake in the 2003 R&D reform. Hence, in [Section 2](#) we estimate the productivity effect of all R&D, including the fake one, and find that it is low. We also diagnose the existence of relabeling in this section and find that it does contribute to the low efficacy of R&D on productivity in China. Therefore, our study provides more direct support for the explanation of relabeling for low R&D efficiency.³⁶

³⁶ Our results also generalize their finding on relabeling because we focus on a reform that covers all firms instead of NHTEs. When we focused on NHTEs only, the estimated effect of R&D on productivity is also quantitatively large, consistent with their estimates.

7. Conclusion

Was China's R&D investment efficient in promoting firm productivity in the period between 2001 and 2007? Our answer is no. In addition to the existing explanations, i.e., distance to the world frontier and misallocation, we provide a third explanation: firms relabel R&D expenses and thus inflate R&D investments. We check this third explanation by examining the effects of the 2003 R&D reform in China.

Innovation plays an important role in promoting economic development. Governments worldwide exert considerable efforts to design and implement CIT policies that can influence firm innovation behaviors. Our study investigates the effect of CIT policy on firm-level innovation, filling the gap in the emerging literature which has focused on the evaluations of R&D-related CIT policy for developing countries. We employ an ownership-based reform of the Chinese CIT additional deduction policy of R&D expenses in late 2003 and the Chinese firm-level dataset from 2001 to 2007 and find that the 2003 R&D reform indeed contributes to improving firm-level R&D investments significantly. The finding is robust to a set of specification checks. Consistent with expectations, the effects are stronger for firms with higher levels of profits and productivity. Although the real effects of the 2003 R&D reform on firm-level productivity are positive and statistically significant, the economic effects of the reform on firm-level productivity are relatively small compared with the existing literature, suggesting that the induced R&D investments are inefficient in improving firm-level productivity.

While existing empirical studies have generally confirmed the efficiency of R&D policies in promoting R&D investment, studies on the real effects of those R&D policies have been scarce. Our study is among a few but still very preliminary. More research along this line is necessary and important.

Acknowledgements

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Appendix

A1. Data process

The original ASIFP data cover approximately 1.93 million observations and 545 thousand firms from 1998 to 2007. We trim the data to fit our analysis and obtain more accurate estimates. Our data are processed by several filters. First, we divide our sample into state-owned sole enterprises, collectively owned enterprises, joint-stock enterprises, and joint-operation enterprises controlled by the state or collective in manufacturing industries, which could enjoy the CIT additional deduction tax policy since 1996. Ownership of the firms is identified based on their legal registration types. We identify the joint-stock and joint-operation enterprises controlled by the state or collective in manufacturing industries if firms have state or collective capital share equal to or greater than 25 %. Second, we drop around 150 observations with negative capital values.

Third, we keep firms that had experienced the 2003 R&D reform, that is, firms that had operation records before 2003 and records after 2003. As a result, around 828 thousand observations in the original sample are removed. Because ownership is an important identification for the 2003 R&D reform, we only keep firms with unchanged ownership status during our sample period. Consequently, around 191 thousand observations have ownerships changed and thus are removed. Moreover, because R&D investment data are not available in 2004, we drop the ASIFP data of 2004, which include around 54 thousand observations.

As a result of the above filtering, we have approximately 80 thousand firms and 346 thousand observations across 478 four-digit manufacturing industries in our sample, as summarized in Table A1.1.

Table A1.1
Number of observations, firms, and industries (by Year).

Year	Observation	Firm	Industry
2001	57,264	57,264	428
2002	75,225	75,225	429
2003	75,480	75,480	473
2005	50,110	50,110	469
2006	45,950	45,950	474
2007	41,862	41,862	473
Total	345,891	79,639	478

A2. The process of estimating productivity

We acknowledge that estimating quantity productivity can be challenging and remains an important research topic, particularly in the field of industrial organization (IO). Due to data limitations, we are unable to directly estimate quantity productivity in our specific

setting. Instead, we estimate revenue productivity based on the available output value of firms.

As the main objective of our paper is to compare our estimated elasticity of R&D on productivity to those in the literature, using revenue productivity allows us to make the comparison more meaningful because most (if not all) of studies in this literature use revenue productivity (e.g., [Aw et al., 2011](#); [Mo et al., 2021](#)). Therefore, we follow [Aw et al. \(2011\)](#)'s paper to estimate revenue productivity, which conveys insights from [Levinsohn and Petrin \(2003\)](#).

Theoretically, the firm in the model is assumed to be a profit maximizer with heterogeneous productivity, which can be derived from the firm's optimal prices based on the marginal cost function and demand elasticity. We assume that the dynamic process of the evolution of productivity follows the Markov process that depends on the firm's R&D investment, exporting participation, and stochastic shocks. The productivity evolution process describes the functional form of this relationship. This specification also allows us to explore the real effect of firms' R&D and exporting activities on productivity empirically.

Specifically, to calculate firm's productivity, we take the following estimation steps:

- (1) By employing the marginal cost function under CES demand assumptions, we obtain optimal market prices and a term covering information of firm-specific productivity and capital stock.
- (2) Based on the information acquired in step (1), the firm's choice of intermediate inputs, and an assumed cubic function capturing the combined effect of capital and productivity on domestic revenue, we derive the value of a term covering the information of demand elasticity, capital stock, and productivity.
- (3) If firms are profit maximizers, total variable cost can be presented by an elasticity-weighted combination of total revenue in the domestic and foreign markets. We use the equation to derive the value of demand elasticity.
- (4) Given the value of demand elasticity, we infer the coefficient of the capital stock in the term we mentioned in step (1).
- (5) Finally, we estimate the firms' productivity given the estimates of the value of demand elasticity, the term in step (1), the coefficient of the capital stock and the observable value of the capital stock.

A3. Data by ownership

[Tables A.3.1, A.3.2, A.3.3](#)

Table A3.1

Requirements of the CIT additional deduction policy.

Requirement	Before the 2003 reform	After the 2003 reform
R&D Growth Rate in two consecutive years	10 %	10 % (canceled in late 2006)
Favored Enterprises	Manufacturing SOEs (110, 141, 143), COEs (120, 142) and their controlled joint-stock enterprises (130) and joint-venture enterprises (149)	Manufacturing Enterprises with all kinds of ownerships (110, 120, 130, 141, 142, 143, 149, 151, 159, 160, 171, 172, 173, 174, 190, 210, 220, 230, 240, 310, 320, 330, 340)
Certifying Agency	Local ministry of S&T Local taxation authority Provincial taxation authority	Local ministry of S&T Local taxation authority Provincial taxation authority

Table A3.2

Domestic firms' registration type in our sample.

Registration type	Firm type	Frequency	Percent
110	State-owned sole enterprises	41,462	11.99
120	Collectively owned enterprises	44,962	13.00
130	Joint-stock enterprises	15,317	4.43
141	State-owned joint-venture enterprises	409	0.12
142	Collective joint-venture enterprises	800	0.23
143	State-owned and collective joint-venture enterprises	1,399	0.40
149	Other joint-venture enterprises	312	0.09
151	State-owned limited corporations	1,781	0.51
159	Other limited liability corporations	60,654	17.54
160	Corporation limited by shares	15,712	4.54
171	Private-owned limited enterprises	40,391	11.68
172	Private-owned partnership enterprises	8,188	2.37
173	Private-owned limited liability corporations	107,148	30.98
174	Private-owned corporation limited by shares	6,749	1.95
190	Other enterprises	607	0.18
Total		345,891	100

Table A3.3

Details of R&D behaviors by year and by group.

Year	Average R&D dummy	Average R&D to sale	(control firms) Average R&D dummy	(control firms) Average R&D to sale	(treated firms) Average R&D dummy	(treated firms) Average R&D to sale
2001	14.38 %	0.140 %	12.19 %	0.120 %	15.51 %	0.151 %
2002	14.50 %	0.146 %	12.35 %	0.124 %	15.35 %	0.154 %
2003	14.72 %	0.146 %	12.14 %	0.125 %	15.76 %	0.155 %
2005	13.76 %	0.202 %	10.77 %	0.166 %	14.60 %	0.213 %
2006	15.10 %	0.248 %	12.45 %	0.218 %	15.78 %	0.256 %
2007	16.49 %	0.289 %	14.24 %	0.264 %	16.98 %	0.294 %

A4. Elasticity

First, we summarize after-tax profit for private firms. We set y as total revenue minus all other costs excluding R&D expenses and set p as after-tax profit.

(1) Before 2003 R&D reform (without CIT additional deduction policy):

- (i) If $y > R\&D$, $p = (y - R\&D) * 67\%$ and CIT is $(y - R\&D) * 33\%$;
- (ii) If $y \leq R\&D$, $p = y - R\&D$ and CIT is 0.

(2) After 2003 R&D reform (with CIT additional deduction policy):

- (a) If $2003 \leq \text{year} < 2006$,
 - (i) If R&D growth rate ≥ 0.1 and $y > 150\% * R\&D$, $p = (y - R\&D) * 67\% + 16.5\% * R\&D$, CIT is $(y - 150\% * R\&D) * 33\%$, and CIT deduction (compared with no CIT additional deduction policy) is $16.5\% * R\&D$;
 - (ii) If R&D growth rate ≥ 0.1 and $150\% * R\&D \geq y > R\&D$, $p = y - R\&D = (y - R\&D) * 67\% + (y - R\&D) * 33\% = (y - R\&D) * 67\% + p * 33\%$, CIT is 0, and CIT deduction is $p * 33\%$;
 - (iii) If R&D growth rate ≥ 0.1 and $y \leq R\&D$, after-tax profit $= y - R\&D$, CIT is 0, and CIT deduction is 0;
 - (iv) If R&D growth rate < 0.1 and $y > R\&D$, after-tax profit $= (y - R\&D) * 67\%$, CIT is $(y - R\&D) * 33\%$, and CIT deduction is 0;
 - (v) If R&D growth rate < 0.1 and $y \leq R\&D$, after-tax profit $= y - R\&D$, CIT is 0, and CIT deduction is 0.
- (b) If $\text{year} \geq 2006$ (R&D growth rate requirement is abolished in 2006),
 - (i) If $y > 150\% * R\&D$, after-tax profit $= (y - R\&D) * 67\% + 16.5\% * R\&D$, CIT is $(y - 150\% * R\&D) * 33\%$, and CIT deduction is $16.5\% * R\&D$;
 - (ii) If $150\% * R\&D \geq y > R\&D$, after-tax profit $= (y - R\&D) * 67\% + \text{after-tax profit} * 33\%$, CIT is 0, and CIT deduction is $p * 33\%$;
 - (iii) If $y \leq R\&D$, after-tax profit $= y - R\&D$, CIT is 0, and CIT deduction is 0.

Next, based on the above equations, we can precisely calculate the decrease in R&D cost and decrease in CIT after the 2003 R&D reform because of the CIT additional deduction policy. If we consider CIT deduction,

$$\ln(RD_{it}) = \beta_0 + \beta_1 \ln(CIT_deduction)_{it} + \gamma X_{it-1} + \lambda_i + \lambda_{jt} + \epsilon_i,$$

where $\ln(CIT_deduction)_{it}$ captures the percentage change in CIT deduction of firm i in year t and the other settings are the same as those in our basic regression. β_1 , the estimated elasticity of R&D, is 0.9147 and statistically significant at 1 % after various firm characteristics and fixed effects are controlled.

A5. Effects of the 2003 R&D reform on other R&D performances

Based on our baseline empirical model (1), we estimate the effect of the 2003 reform on firm-level R&D intensity (R&D investments/sales) in column (1) and R&D participation in column (2). We further estimate the policy effect on firm-level R&D investments using a subsample of the firms with positive pre-2003 R&D in column (3). Results are summarized in [Table A5.1](#).

Table A5.1
Baseline results.

Variable	(1) R&D Intensity	(2) R&D Dummy	(3) $\ln(R\&D)$ (Positive)
$D_t * Post03_t$	0.0005*** (0.0002)	0.0033 (0.0038)	0.1300* (0.0741)

(continued on next page)

Table A5.1 (continued)

Variable	(1) <i>R&D Intensity</i>	(2) <i>R&D Dummy</i>	(3) <i>ln(R&D)</i> <i>(Positive)</i>
$\ln(\text{labor})_{it-1}$	0.0002 (0.0001)	0.0315*** (0.0027)	0.4900*** (0.0438)
$\ln(\text{kl})_{it-1}$	0.0000 (0.0001)	0.0051*** (0.0013)	0.0783*** (0.0235)
$\text{foreignShare}_{it-1}$	−0.0006 (0.0006)	−0.0188 (0.0261)	−0.1328 (0.5441)
tfp_{it-1}	−0.0000 (0.0001)	0.0097*** (0.0014)	0.2132*** (0.0251)
$\ln(\text{age})_{it-1}$	0.0001 (0.0002)	−0.0053 (0.0061)	−0.3074*** (0.1081)
$\ln(\text{age}^2)_{it-1}$	−0.0000 (0.0000)	0.0014 (0.0016)	0.0589** (0.0248)
$\ln(\text{export})_{it-1}$	0.0000*** (0.0000)	0.0013*** (0.0004)	0.0261*** (0.0067)
RegType_{it-1}	Y	Y	Y
Industry-Year FE	Y	Y	Y
Firm FE	Y	Y	Y
Observations	263,218	263,272	66,302
R-Square	0.480	0.631	0.627

Note: Registration type_{it-1} includes a set of dummy variables indicating the firm's registered type in year t-1. We do not report the coefficients to save space. *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors are clustered at the industry-ownership level.

A6. Effects of the 2003 R&D reform based on alternative comparable groups

We summarize the matching regression results and the balance property test results based on the PSM method in this subsection. [Tables A.6.1, A.6.2](#)

Table A6.1
Matching regression results (Logit).

Variable	<i>Treatment (D_t)</i>
Labor	−0.0002702*** (0.0000222)
Capital-labor ratio	0.0017876*** (0.0001509)
Foreign share	−0.3195915 (0.2265031)
Total sale	0.00000152*** (0.000000143)
Total asset	−0.000000317*** (0.000000078)
Observations	55,911

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

Table A6.2
Balance property test results.

Variable	(1) <i>Treated</i>	(2) <i>Control</i>	(3) <i>Difference</i>
Labor	275.87	279.13	3.26 (0.721)
Capital-labor ratio	46.724	47.635	0.911 (0.184)
Foreign share	0.00279	0.00258	−0.00021 (0.449)
Total sale	49,518	50,090	572 (0.828)
Total asset	58,545	64,938	6,393 (0.134)

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. P-values are in the brackets.

A7. Direct Effect of the 2003 R&D Reform on Firm-level Productivity

We also examine the direct effect of the 2003 R&D reform on firm-level productivity using the DID method. In particular, we employ the Eq. (1) but replace the firm-level R&D expenses by firm-level productivity. Productivity is measured based on OP, LP, OP-ACF, and LP-ACF methods according to Olley and Pakes (1996), Levinsohn and Petrin (2003), and Akerberg et al. (2015), respectively. Results are presented in Table A7.1.

The coefficients of the DID term are all significant and positive, suggesting a positive effect of the reform on the productivity. However, given the average values of productivity are 2.4128 (OP), 6.5295 (LP), 2.3284 (OP-ACF), and 3.5832 (LP-ACF) in the pre-reform period, the percent increase of the productivity induced by the reform is small.

Table A7.1
Productivity Effect of the Reform.

Variable	OP (1) TFP	LP (2) TFP	OP-ACF (3) TFP	LP-ACF (4) TFP
$D_t \cdot \text{Post03}_t$	0.0547*** (0.0115)	0.1086*** (0.0138)	0.0595*** (0.0127)	0.1056*** (0.0144)
Firm controls	Y	Y	Y	Y
Firm FE	Y	Y	Y	Y
Industry-Year FE	Y	Y	Y	Y
Observations	183,107	183,107	183,107	183,107
R-Square	0.910	0.872	0.902	0.973

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$. Standard errors are clustered at the industry-ownership level.

A8. Alternative specifications with a different set of controls

We explore alternative specifications with a different set of controls to confirm our estimations. Specifically, in column (1) in Table A8.1, we include firms' past importing activity to control for the impacts of imported technology and inputs. In column (2), we add industry-year fixed effects to control for industry-specific trends and other industrial policies such as tariff changes and FIE spillovers, and city-year fixed effects to take regional reforms into account such as the VAT tax reform. The estimation results suggest that our main finding is not sensitive to these specifications. We also find that further including the ownership dummy in the model using the full sample, as shown in column (3), does not change the pattern of our finding. In addition, we rely on the subsample of treated firms (mostly private firms) to explore the R&D efficiency on the mechanisms, which is immune from the SOE concerns.

In column (4), we employ the two-stage IV regression method and instrument the past R&D activity in the productivity evolution process using the 2003 R&D reform to investigate the effects of R&D investments on firms' productivity. The IV estimate is significantly positive and also small, which is consistent with our main finding based on the OLS regression that the R&D efficiency is relatively small. We note that the IV estimate is larger than the OLS estimate. This is possible because the IV method estimates the local average effect while the OLS method estimates the average treatment effect over the entire sample. In our setting, the IV method mainly focuses on the firms influenced by the 2003 reform and how the policy-induced R&D investments affect the productivity evolution, while the OLS method estimates the average effects of all firms' R&D investments on productivity.

We further add firms' past importing, industry-year fixed effects, and city-year fixed effects in column (5). The estimate is still significant and positive. To address the concern that R&D today may also affect R&D tomorrow and therefore R&D investments become serially correlated, we include the one-period lag R&D investments in our first stage regression in column (6). The pattern of the estimate in second-stage regression does not change.

Table A8.1
Results with different controls and under IV method.

Variable	(1) Subsample Treated firms Productivity	(2) Subsample Treated firms Productivity	(3) Subsample All firms Productivity	(4) Subsample Treated firms (Two-stage IV) Productivity	(5) Subsample Treated firms (Two-stage IV) Productivity	(6) Subsample Treated firms (Two-stage IV) Productivity
α_4	0.00000329*** (0.000000784)	0.00000616*** (0.000000795)	0.00000753*** (0.000000727)	0.0000685*** (0.00000643)	0.0001392*** (0.00000772)	0.0001396*** (0.00000796)
α_5	−0.0000168*** (0.00000341)	0.00000934** (0.00000366)	0.00000476 (0.00000337)	−0.0000178* (0.00000992)	0.0000209* (0.0000109)	0.0000302*** (0.0000105)
α_6	0.00000489*** (0.00000120)	0.00000375*** (0.00000119)	0.00000110 (0.00000111)	0.00000643 (0.00000903)	−0.0000244** (0.00000977)	−0.0000323*** (0.00000971)

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Table A8.1 (continued)

	(1) Subsample Treated firms	(2) Subsample Treated firms	(3) Subsample All firms	(4) Subsample Treated firms (Two-stage IV)	(5) Subsample Treated firms (Two-stage IV)	(6) Subsample Treated firms (Two-stage IV)
Variable	<i>Productivity</i>	<i>Productivity</i>	<i>Productivity</i>	<i>Productivity</i>	<i>Productivity</i>	<i>Productivity</i>
Import activity	Y	Y	Y	N	Y	Y
SOE dummy	N	N	Y	N	N	N
Industry-Year FE	N	Y	Y	N	Y	Y
City-Year FE	N	Y	Y	N	Y	Y
Previous R&D	N	N	N	N	N	Y
Observations	173,253	173,253	231,002	132,928	132,928	132,928

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

A9. Relabeling results based on regression discontinuity design method

We check the results using Regression Discontinuity Design (RDD) regressions with various combinations of bandwidths (i.e., optimal bandwidth, half of the optimal bandwidth, and twice of the optimal bandwidth) and degrees of polynomial functions (i.e., 2nd polynomial and 3rd polynomial functions). The estimates are reported in the following Table A9.1. We do not find significant discontinuity in firms' R&D to administration expense ratio and non-R&D to sale ratio at the threshold in 2002, but find significant discontinuities of these ratios at the threshold in 2006. Some estimates under twice optimal bandwidths are insignificant but consistent in sign. A possible reason is that the large bandwidth covers too many firms that are far away from the threshold, so the estimates may not be precisely estimated. Meanwhile, the jumps around the threshold in the log sales are all insignificant under various combinations of bandwidths and degrees of polynomial functions. Nevertheless, the results in 2002 and 2006 imply firms' relabeling activities, which are consistent with Figs. 4 and 5.

Table A9.1

Results based on RDD (R&D to adm expense ratio and Non-R&D to sales ratio).

Variable	2002 (1) R&D to adm expense ratio (2nd poly)	2006 (2) R&D to adm expense ratio (2nd poly)	2002 (3) Non-R&D to sales ratio (2nd poly)	2006 (4) Non-R&D to sales ratio (2nd poly)	2002 (5) R&D to adm expense ratio (3rd poly)	2006 (6) R&D to adm expense ratio (3rd poly)	2002 (7) Non-R&D to sales ratio (3rd poly)	2006 (8) Non-R&D to sales ratio (3rd poly)
Discontinuity (optimal bwd)	−0.0134 (0.0450)	0.3467** (0.1707)	0.0073 (0.0410)	−0.0146* (0.0094)	−0.0270 (0.0503)	0.3628** (0.1887)	0.0010 (0.0489)	−0.0153* (0.0090)
Discontinuity (0.5optimal bwd)	−0.0049 (0.0495)	0.3348* (0.1933)	−0.0015 (0.0518)	−0.0152* (0.0087)	−0.0038 (0.0504)	0.4429** (0.2140)	0.0117 (0.0589)	−0.0179** (0.0093)
Discontinuity (2optimal bwd)	0.0798 (0.0456)	0.1906 (0.1419)	−0.0062 (0.0279)	−0.0091 (0.0097)	0.0793 (0.0481)	0.3284** (0.1674)	−0.0228 (0.0386)	−0.0135 (0.0097)

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

Table A9.2

Results based on RDD (log sales).

Variable	2002 (1) <i>ln(sale)</i> (2nd poly)	2006 (2) <i>ln(sale)</i> (2nd poly)	2002 (3) <i>ln(sale)</i> (3rd poly)	2006 (4) <i>ln(sale)</i> (3rd poly)
Discontinuity (optimal bwd)	−0.3117 (0.8572)	0.3572 (0.2669)	−0.2889 (0.8729)	0.4302 (0.2825)
Discontinuity (0.5optimal bwd)	0.2753 (0.8813)	0.4738 (0.2925)	0.2720 (0.8958)	0.5794* (0.3281)
Discontinuity (2optimal bwd)	0.6945 (0.7120)	0.0642 (0.2414)	0.5484 (0.7337)	0.2805 (0.2637)

Note: *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.1$.

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